Ethan Cox

(816)-679-8121 | ethancox@iastate.edu | github.com/ethancox-isu | Secret Clearance

Experience

AI Research Assistant Aug. 2026 – Present

Software Analytics and Pervasive Parallelism Lab (SwAPP) – Iowa State University, Intel Labs *Ames, IA*

- Training heterogeneous GNNs in PyTorch/DGL (RGCN with HeteroGraphConv) with a numerically aware MLIR program representation of C/C++ code that adds aggregate-data-type nodes and a digit-and-position embedding so the model handles numbers unseen in training.
- Achieved 6% accuracy on AMD and 10% on NVIDIA for the CPU/GPU device mapping challenge on 680 LLVM IR instances from 256 OpenCL kernels with 10-fold cross-validation.

Software Engineer CO-OP Jun. 2026 – Present

Johns Hopkins Applied Physics Laboratory – Signals Exploitation Systems Group *Laurel, MD*

- Developed an RF geolocation engine in object-oriented C++ & MATLAB for a special operations targeting system (RICO) with production code shipping directly into the program's Critical Design Review (CDR).
- Implemented an Unscented Gaussian Mixture Filter algorithm as a probabilistic baseline for nonlinear state estimation of RF-emitting transmitters with signal degradation sources, including stochastic waveform noise.
- Built a C++ MEX-CUDA kernel interface for MATLAB that offloads intensive convolution-based interpolation onto a parallelized NVIDIA GPU to dramatically reduce processing time for multiplexed aircraft trajectory signals.
- Established a GitLab CI/CD pipeline to automate testing across system-on-module (SOM) targets.

Software Engineer CO-OP Jul. 2025 – Dec. 2025

MIT Lincoln Laboratory – Systems Engineering Group *Lexington, MA*

- Engineered a full-stack, on-premise multi-agent AI platform that validates MBSE systems models in an air-gapped environment with a locally hosted Mistral 7B SLM, chosen for its size and benchmark performance to keep infrastructure cost and latency low while retaining semantic reasoning capability. Deployed on a supercomputing center and verified on a standard laptop via Ollama, for air-gapped environments where cloud AI is restricted.
- Built the platform's stack (React frontend, FastAPI backend, Neo4j graph database, LlamaIndex agent layer) and a Python GraphRAG pipeline that converts MBSE textual notation into ontology-based knowledge graphs and retrieves only the relevant subgraph per query, avoiding SLM context-window limits and reducing compute.
- Co-authored a research paper benchmarking the agents against four MIT LL systems engineers on a 12-requirement CubeSat model, where the agents were 10x faster than the fastest engineer (0.25 vs. 2.75 min) with 86% average correctness vs. 77% for engineers, and surfaced a missing SSD part even a senior engineer (10+ years) overlooked.

Software Engineer Intern (x2) May 2023 – Aug. 2023, May 2024 – Aug. 2024

EverCommerce *Virtual*

- Designed and shipped Python REST APIs for a production LLM-powered AI Copilot serving real users, handling transcription, structured data extraction, and EHR system integration at scale across a live healthcare platform.
- Built and deployed a production ETL pipeline on AWS Redshift using Fivetran and dbt, transforming raw operational data into standardized business metrics consumed by managers across the organization.

Education

Iowa State University Ames, IA

B.S. in Computer Science, Minor in Applied Math – 3.83 GPA *Expected May 2028*

Awards: Computer Science Excellence Award, Academic Achievement Award, 2026 Student Spotlight Award

Coursework: Software Development, Advanced Programming, Algorithms, Data Structures, Linear Algebra

Publications

A Multi-Agent Approach to Validating Systems Models 2026

- Research paper accepted in International Council on Systems Engineering (INCOSE) - International Symposium.
- Co-authored with MIT Lincoln Laboratory and MIT Lincoln Laboratory Supercomputing Center (LLSC).

Skills

Programming: Python, C/C++, MATLAB, Java, SQL, Bash

Systems & Infra: Linux, Docker, Git/GitLab CI/CD, AWS, GPU computing (CUDA)

AI/ML: PyTorch, TensorFlow, LLMs, RAG/GraphRAG, NLP, agentic AI (LangGraph, CrewAI, Autogen)

Concepts: High-performance computing, benchmarking, system validation, distributed/cluster computing